

GATE QUESTION Q36

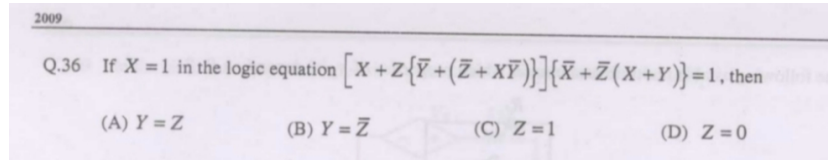


Figure 1: GATE Question Q36

Contents

1	Components	2
1.1	Breadboard	2
1.2	IC 7474	2
1.3	IC 7447	2
1.4	Arduino Uno	2
2	Boolean Simplification	2
3	Hardware Connections	2
3.1	Arduino and 7474 Connections	2
3.2	Arduino to 7447 Connections	3
4	Software Implementation	3
5	Output	4
6	Result	5

1 Components

Component	Specification	Quantity
Breadboard	Standard	1
Arduino Uno	ATmega328P	1
IC 7474	D Flip-Flop	2
IC 7447	BCD to 7 Segment Decoder	1
Seven Segment Display	Common Anode	1
Jumper Wires	Male-Male	As Required

1.1 Breadboard

The breadboard is used for temporary circuit connections.

1.2 IC 7474

The 7474 is a dual D Flip-Flop IC used to store binary data.

1.3 IC 7447

The 7447 is a BCD to Seven Segment Decoder Driver.

1.4 Arduino Uno

Arduino Uno is used to generate the clock signal and read the outputs from the 7474 ICs.

2 Boolean Simplification

For $X=1$,
[$F=$]
For output 1,
[$Z=0$]

3 Hardware Connections

3.1 Arduino and 7474 Connections

Arduino Pin	Function heightD13
Clock D6	W (Q1 of IC1) D7
X (Q2 of IC1) D8	Y (Q1 of IC2) D9
Z (Q2 of IC2) height	

3.2 Arduino to 7447 Connections

Arduino Pin	7447 Pin heightD2
A D3	B D4
C D5	D height

4 Software Implementation

```
#include <Arduino.h>
```

```
int Z;  
int D,C,B,A;
```

```
void disp_7447(int D,int C,int B,int A)  
{  
  digitalWrite(2,A);  
  digitalWrite(3,B);  
  digitalWrite(4,C);  
  digitalWrite(5,D);  
}
```

```
void setup()  
{  
  pinMode(2,OUTPUT);  
  pinMode(3,OUTPUT);  
  pinMode(4,OUTPUT);  
  pinMode(5,OUTPUT);
```

```
  pinMode(6,INPUT);  
  pinMode(7,INPUT);  
  pinMode(8,INPUT);  
  pinMode(9,INPUT);
```

```

pinMode(13,OUTPUT);
}

void loop()
{
digitalWrite(13,HIGH);
delay(10);
digitalWrite(13,LOW);

Z = digitalRead(9);

int F = !Z;

if(F==0)
{
D=0; C=0; B=0; A=0;
}
else
{
D=0; C=0; B=0; A=1;
}

disp_7447(D,C,B,A);
delay(1000);
}

```

5 Output

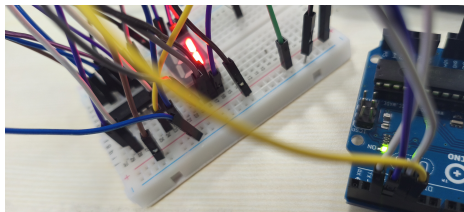


Figure 2: Output Showing 1

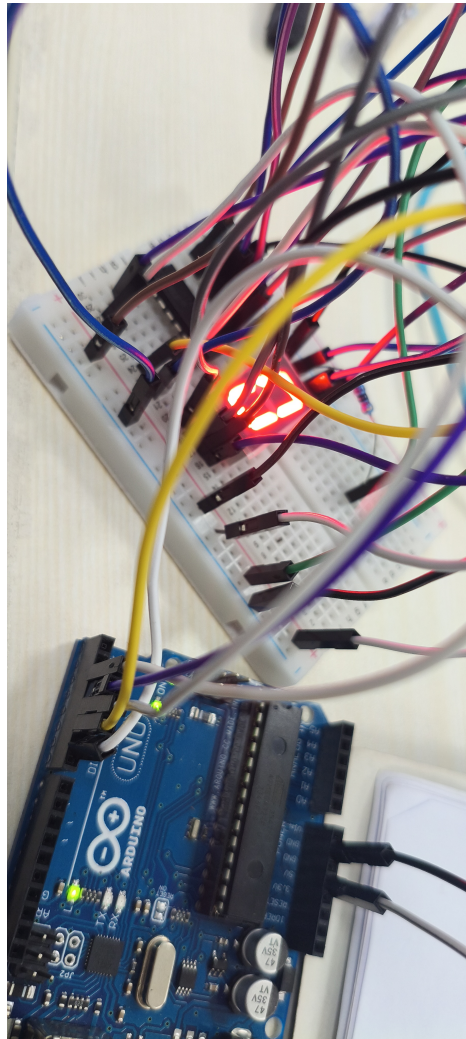


Figure 3: Output Showing 0

6 Result

The given Boolean expression was implemented using 7474 D Flip-Flops, 7447 Decoder and a Seven Segment Display.

[F=]

When Z=0,

[F=1]

Hence the display shows 1 on the Seven Segment Display.